



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Software Engineering Iteration 0

The Smart Retail Demand Forecasting System

(RetailIQ)

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Smart Retail Demand Forecasting System

(RetailIQ)

1. Project Overview

RetailIQ is a web-based retail analytics platform designed to help businesses forecast product demand and understand the factors influencing sales performance. The system uses historical retail sales data to generate demand forecasts and provide meaningful insights through interactive dashboards and analytical tools.

Traditional retail forecasting tools only predict future sales numbers but fail to explain the reasons behind demand fluctuations. RetailIQ improves this by providing interpretable analysis of sales patterns, promotion impacts, and product relationships. The platform allows users to upload sales datasets, visualize sales trends, analyze promotional effects, and receive business recommendations based on data-driven insights.

The system integrates machine learning models with a user-friendly web interface to help retail managers make informed decisions regarding inventory management, promotions, and product stocking strategies. Through interactive visualizations and analytical dashboards, RetailIQ transforms complex retail data into actionable insights.

2. Problem Statement

Retail businesses often struggle to accurately forecast product demand and understand the factors that influence sales changes. Incorrect demand predictions can result in stock shortages, lost revenue opportunities, or excessive inventory that increases operational costs.

Most traditional forecasting systems provide only total sales predictions without explaining the drivers behind those predictions. Retail managers therefore lack visibility into whether sales changes are caused by seasonal trends, promotional campaigns, or shifts in demand between related products.

This lack of transparency limits effective decision-making in areas such as promotion planning, product stocking, and inventory optimization. Therefore, a system is needed that not only forecasts demand but also explains the underlying sales behavior in a clear and interpretable manner.

3. Problem Solution

RetailIQ provides a smart web-based solution that combines demand forecasting with data visualization and retail analytics. The system analyzes historical retail sales data to generate forecasts while also identifying patterns and factors influencing product demand.

The platform provides dashboards that visualize sales trends, promotional effects, and top-performing products. It also enables users to filter sales data by time periods, analyze promotional impacts, and receive recommendations for improving retail performance.

Additionally, the system allows administrators to communicate important system updates and announcements to users, while users can provide feedback and access helpful documentation through the platform.

By integrating machine learning analysis with an interactive web interface, RetailIQ enables retail managers to better understand sales behavior and make more informed strategic decisions.

4. Scope of the Project

The scope of this project focuses on developing a web-based analytics platform capable of forecasting retail product demand and providing business insights through visual dashboards.

The system will allow users to upload historical sales datasets, explore sales trends, analyze promotional impacts, and generate demand forecasts using regression-based machine learning models. It will also include interactive dashboards, business recommendations, and analytical reports to support retail decision-making.

The project will focus on building a complete analytical platform with visualization and forecasting capabilities rather than integrating with real-time retail enterprise systems. Deep learning techniques will not be used; instead, the project emphasizes interpretable statistical and machine learning approaches suitable for business analytics.

5. Basic Features

Feature	Description
User Authentication System	Allows users to securely register and log in to the system. This ensures that only authorized users can access forecasting tools and dashboards.
Dashboard Overview	Provides a quick summary of key retail metrics such as total sales, number of products, promotion statistics, and forecast summaries using visual cards and charts.
Sales Trend Visualization	Displays historical sales patterns through line charts and graphs showing daily, weekly, or monthly sales trends to help users identify seasonality and demand changes.
Promotion Impact Analysis	Shows how promotions influence sales by comparing normal sales with promotional sales periods. This helps users understand whether promotions truly increase demand.
Time Filter Option	Allows users to filter sales data by different time ranges such as weekly, monthly, or yearly. This makes analysis more flexible and interactive.
Top Performing Products Page	Displays the top five best-selling products based on sales or forecast growth using ranking tables and charts.
Recommendations Page	Provides business recommendations generated from the system's analysis, such as suggesting products that should be stocked more during certain seasons.
Announcements Page	Allows administrators to post system updates such as new dataset uploads, model updates, or important system announcements for users.
News & Insights Page	Displays short articles or tips about retail trends, demand forecasting strategies, and promotional insights to enhance the professional look of the platform.
Feedback Page	Allows users to submit feedback, suggestions, or comments about forecasts and system functionality, which are stored in the database for review.
Help / User Guide Page	Provides instructions on how to use the dashboard, forecasting tools, and analysis features to make the system easier to understand for new users.

6. Project Plan

Iteration Plan (Agile Development)

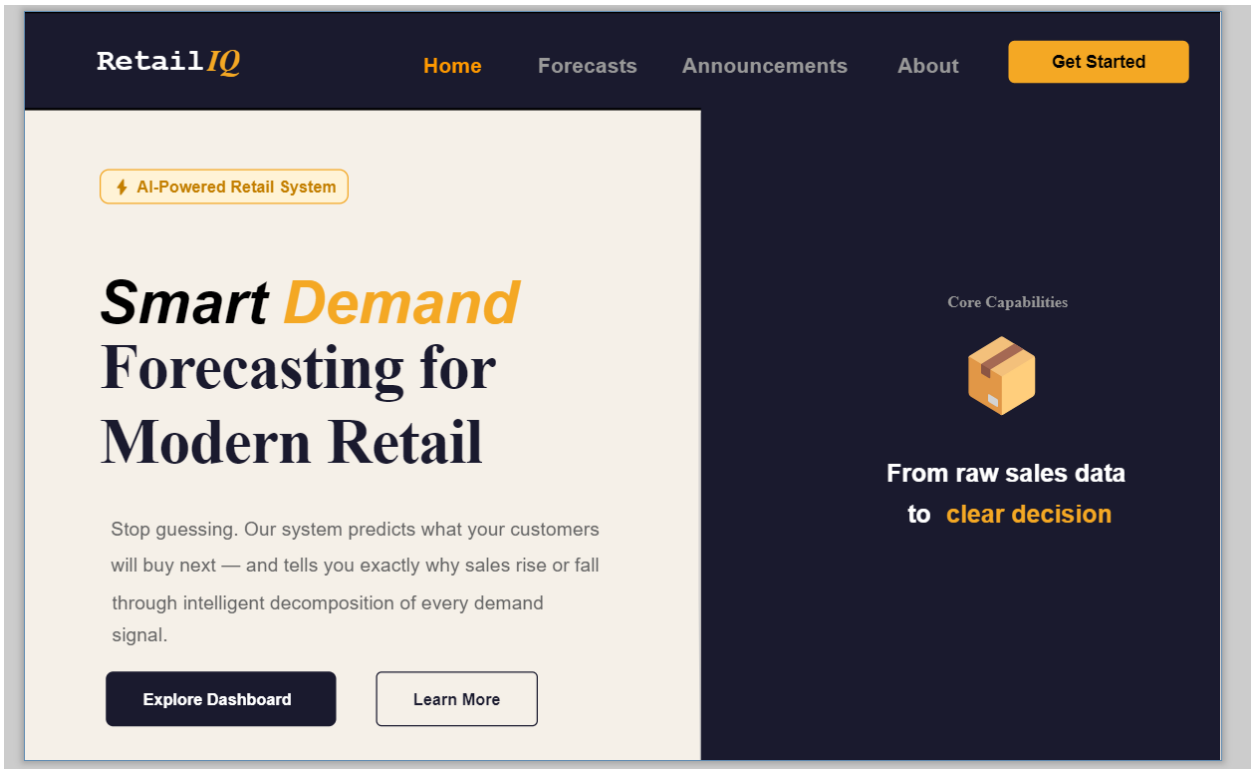
Iteration	Duration	Modules Developed	User Stories
Iteration 1	Weeks 1–2	Authentication, Announcements Page, Dataset Upload, Basic Dashboard	User can register/login, upload sales dataset, and view basic system announcements and sales overview
Iteration 2	Weeks 3–4	Sales Trend Visualization, Time Filters, Promotion Impact Analysis, Top Products Page	User can analyze sales trends, apply time filters, and view top-performing products
Iteration 3	Weeks 5–6	Recommendations Page, News & Insights, Feedback System, Help Page, Forecast Integration	Users can view business recommendations, read insights, submit feedback, and use the forecasting system

Team Roles

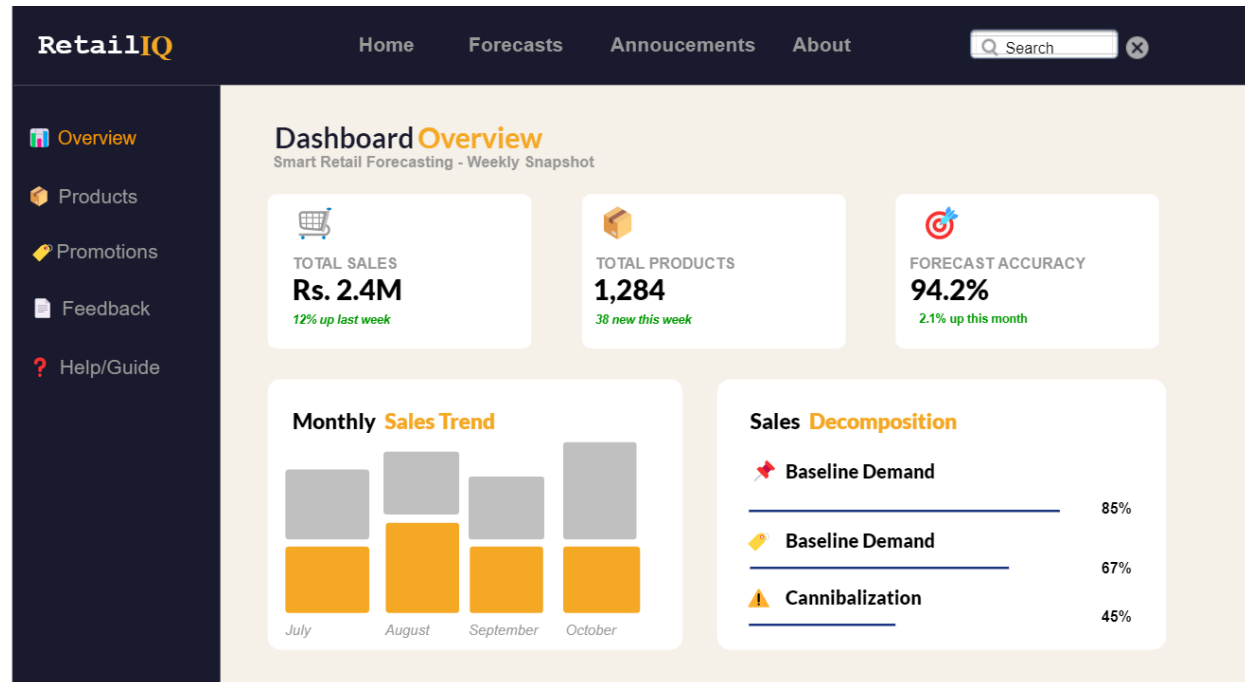
Name	Role
Ayesha Khalid	Project Lead, Frontend Developer, Machine Learning Engineer
Uma E Rubab	Backend Developer, System Architect, Tester & Quality Assurance

Prototype

Home Page:



Dashboard:



Announcements Page:

The screenshot shows the 'Announcements' page of the RetailIQ system. The header includes the RetailIQ logo, navigation links (Home, Forecasts, Announcements, About), and a search bar. A left sidebar contains links to Overview, Products, Promotions, Feedback, and Help/Guide. The main content area is titled 'System Announcements' with a subtitle 'Latest Updated from Admin - Notice Board'. It features three announcement cards: 1. 'New Dataset Uploaded --- Rossman Store Sales December 2024' with a 'Dataset' button. 2. 'New Forecast Available --- Week of Dec 2 to Dec 8' with a 'Forecast' button. 3. 'Demand Model Updated -- L2 Regularization Improved' with a 'Model' button.

RetailIQ Home Forecasts **Announcements** About

System Announcements
Latest Updated from Admin - Notice Board

New Dataset Uploaded --- Rossman Store Sales December 2024 Dataset
A new retail sales dataset have been added to the system. Contains 6 months of product-level data across 50 stores. Models will be retrained automatically.

New Forecast Available --- Week of Dec 2 to Dec 8 Forecast
Weekly demand forecasts for all 1,284 products are new ready. Overall forecast accuracy standard stands at 92.4%. Visit forecast page to see details.

Demand Model Updated -- L2 Regularization Improved Model
The regression model have been trained with L2 regularization parameters. Overfitting reduced by 8%. New model is now live for all forecasts.

Help/User Guide Page:

The screenshot shows the 'Help & User Guide' page of the RetailIQ system. The header and sidebar are identical to the Announcements page. The main content area is titled 'Help & User Guide' with a subtitle 'Step-by-step instructions on how to use RetailIQ' and a search bar 'Search guide topics...'. It features six numbered steps in a grid: 1. 'Open Home Page' (house icon), 2. 'View Dashboard' (bar chart icon), 3. 'View Promotions' (tag icon), 4. 'Check Announcements' (megaphone icon), 5. 'View Products' (box icon), and 6. 'Write Feedback' (document icon).

RetailIQ Home Forecasts Announcements About

Help & User Guide
Step-by-step instructions on how to use RetailIQ

Step 01 **Open Home Page**
Start from the Home page. Read the and click "Explore Dashboard" to enter the system

Step 02 **View Dashboard**
See total sales, products, promotions, and forecast accuracy all in one overview screen.

Step 03 **View Promotions**
Visit the Promotions page to view all active discounts and check which deals are genuinely boosting your overall sales.

Step 04 **Check Announcements**
Check the Announcements page regularly for dataset updates, model changes, and new forecasts.

Step 05 **View Products**
Browse and manage all tracked products available in the retail system.

Step 06 **Write Feedback**
Share your thoughts or report any issue to help improve the system